SOUMIK MUKHOPADHYAY

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RESEARCH INTEREST

My research interests lie in computer vision, particularly at the intersection of generative modeling, multimodal learning, representation learning, and efficient training. I study how structure across scales, learned representations, and generative dynamics can improve visual understanding and enable efficient image and video generation. I have prior experience in medical image analysis.

EXPERIENCE

UNIVERSITY OF MARYLAND, COLLEGE PARK | RESEARCH ASSISTANT — May 2021 - Present
Forgery detection, Sparse 3D novel view synthesis — Prof. Abhinav Shrivastava | Dept. of Computer Science

ADOBE, SAN JOSE | RESEARCH SCIENTIST INTERN — May 2026 - Present
Pixel diffusion models — Aniruddha Mahapatra, Aseem Agarwala | Video AI Lab

ADOBE, SAN JOSE | RESEARCH SCIENTIST INTERN — May 2024 - Nov 2024
Multimodal generative models — Dr. Eli Shechtman | Creative Intelligence Lab

PRIME VIDEOS & AMAZON STUDIOS, CULVER CITY | APPLIED SCIENTIST INTERN — May 2022 - Aug 2022
Conditional video generative models — Naveen Nair, Ravi Teja Gadde | Applied Science

SAMSUNG R&D INSTITUTE BANGALORE | SOFTWARE ENGINEER (RESEARCH) — June 2018 - Aug 2020
Medical image analysis (prostate fusion biopsy) — Bhavya Ajani | Health and Medical Equipment

SAMSUNG R&D INSTITUTE BANGALORE | INDUSTRIAL INTERN — May 2017 - July 2017
Speech assistant's large utterance optimization — Vrajesh Sejpal | Voice Intelligence

INDIAN STATISTICAL INSTITUTE, KOLKATA| RESEARCH INTERN — May 2016 – Aug 2016
Quantum algorithms (k-SUM problem) — Prof. Subhamoy Maitra | Applied Statistics Unit

EDUCATION

UNIVERSITY OF MARYLAND, COLLEGE PARK
Aug 2020 - Present | College Park, MD, USA
PH.D. IN COMPUTER SCIENCE (VISUAL GENERATIVE MODELS)
Department of Computer Science
Advisor - Prof. Abhinav Shrivastava | GPA: 4.0 / 4.0

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI
June 2014 - May 2018 | Guwahati, India
BTECH IN COMPUTER SCIENCE AND ENGINEERING
Department of Computer Science and Engineering
Department Rank 5 (out of 88) | GPA: 9.38 / 10.0

PAPERS

Rethinking the Design Space of Vanilla Pixel Diffusion Transformers
Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra| Under Submission, 2026
Adaptive prediction parameterization and lightweight high-frequency refinement posttraining to improve Just Image Transformer (JiT).

Scale Space Diffusion
Soumik Mukhopadhyay, Prateksha Udhayanan, Abhinav Shrivastava| CVPR, 2026
Why process noisy information-poor states at full resolution? Infusing scale spaces into diffusion for efficient pixel space diffusion.

Selective Timestep Weighting and Advantage-Based Replay for Sample-Efficient Diffusion RLHF
Eric Zhu, Abhinav Shrivastava**, Soumik Mukhopadhyay**| CVPRW (GCV, CVEU), 2026
Emphasizing information diffusion timesteps and trajectories during diffusion RLHF training provides up to 6× sample efficiency.

Do text-free diffusion models learn discriminative visual representations?
Soumik Mukhopadhyay, Matthew Gwilliam, Yosuke Yamaguchi, Vatsal Agarwal, Namitha Padmanabhan, Archana Swaminathan, Tianyi Zhou, Jun Ohya, Abhinav Shrivastava| ECCV, 2024
Diffusion models are self-supervised representation learners that can be used for various discriminative tasks as well.

Diffusion Models Beat GANs on Image Classification
Soumik Mukhopadhyay, Matthew Gwilliam, Yosuke Yamaguchi, Vatsal Agarwal, Namitha Padmanabhan, Archana Swaminathan, Srinidhi Hegde, Tianyi Zhou, Abhinav Shrivastava| Preprint only, 2023
Unconditional diffusion models' representations can be used for classification.

Diff2Lip: Audio Conditioned Diffusion Models for Lip-Synchronization
Soumik Mukhopadhyay, Saksham Suri, Ravi Teja Gadde, Abhinav Shrivastava| WACV, 2024
Diffusion models to synchronize face videos with arbitrary audio.

Deep learning based needle tracking in prostate fusion biopsy
Soumik Mukhopadhyay, Praful Mathur, Aditya Bharadwaj, Yuri Son, Jun-Sung Park, Srinivas Rao Kudavelly, Sangha Song, Hokyung Kang | SPIE Medical Imaging, 2021 (*Best Poster Award in Image-Guided Procedures, Robotic Interventions, and Modeling)*
Real-time localization of needle in an Ultrasound video of Prostate Biopsy and mapping to the 3D MRI space for analysis.

Rigid and deformable corrections in real-time using deep learning for prostate fusion biopsy
Aditya Bhardwaj, Jun-Sung Park, **Soumik Mukhopadhyay**, Sikander Sharda, Yuri Son, Bhavya Ajani, Srinivas Rao Kudavelly | SPIE Medical Imaging, 2020
Real-time MR-TRUS prostate registration: DL-based segmentation, elastix rigid registration, DL-based deformable correction.

PATENT

Methods and biopsy systems of detecting position of a needle during a biopsy procedure
Aditya Bharadwaj, **Soumik Mukhopadhyay**, Junsung Park, Srinivas Rao Kudavelly, Yuri Son
| Inventor on Indian Patent No. 572418 *(Granted 2025)*

[1] *University of Maryland, College Park,* [2] *Samsung R&D Institute Bangalore,* [3] *Indian Institute of Technology Guwahati,* [4] *Indian Statistical Institute Kolkata*
[5] *Prime Videos & Amazon Studios,* [6] *Adobe*

OTHER PROJECTS

1. Research Projects

Adobe[6]

MULTIMODAL GENERATION

AUDIO-VISUAL JOINT GENERATION USING COMBINED TOKENIZATION — May 2024 - Nov 2024
Dr. Prem Seetharaman, Dr. Eli Shechtman, Adobe

UMD[1]

FEW-VIEW 3D RECONSTRUCTION

SINGLE VIEW AUGMENTATION FOR IMPROVING DIFFUSION CLEANUP OF FEW-VIEW 3D — Sept 2025 - May 2026
Prof. Matthias Zwicker, Prof. Abhinav Shrivastava, Prof. Tom Goldstein, Dept. of Computer Science, University of Maryland

UNIFIED GENERATIVE REPRESENTATION LEARNING

GENERATIVE SELF-CONDITIONING FOR IMPROVED GENERATION — Apr 2024 - Nov 2024
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

VIDEO GENERATIVE MODELS

LIP-SYNCHRONIZATION IN HIGH-DEFINITION USING DIFFUSION MODELS — Sept 2023 - Aug 2024
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

IMPLICIT NEURAL REPRESENTATION

GENERATION IN NEURAL REPRESENTATION FOR VIDEOS (NeRV) SPACE — Feb 2023 - Nov 2023
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

MISCELLANEOUS

SPARSE FLOW BASED IN-BETWEENING FOR LINE DRAWINGS IN ANIMATION — Aug 2022 - Oct 2022
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

GENERALISED NEAREST NEIGHBOUR-BASED IMAGE GENERATION — Feb 2022 - May 2022
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

FACIAL PRIVACY IN GAN GENERATED IMAGES — Mar 2021 - Dec 2021
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

DETECTION OF FACIAL FORGEY USING GRAPH NEURAL NETWORKS — Mar 2021 - Aug 2021
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland

SRIB[2]

MEDICAL IMAGE ANALYSIS

FETAL SKULL PREDICTIVE DETECTION/DATA AUGMENTATION USING DCGAN — Nov 2018 – Feb 2018
Nitin Singhal, Senior Chief Engineer, Samsung R&D Institute Bangalore

FOLLICLE DETECTION (FASTER-RCNN) & SEGMENTATION (UNET-BiLSTM) — Sept 2018 – Nov 2018
Nitin Singhal, Senior Chief Engineer, Samsung R&D Institute Bangalore

IITG[3]

COMPUTER VISION

END-TO-END DFT-BASED IMAGE DE-FENCING — Oct 2017 – Feb 2018
Prof. Arijit Sur, Associate Professor, Dept. of CSE, IIT Guwahati

2. Course Projects

UMD[1]

VISION+TEXT

FINE-GRAINED IMAGE-TO-IMAGE EDITING FROM TEXT CAPTIONS — Sept 2020 - Dec 2020
Prof. Abhinav Shrivastava, Dept. of Computer Science, University of Maryland
StyleGAN2 style attribute + BERT encoding for unsupervised image editing.

CONTINUAL LEARNING

ADVERSARIAL ROBUSTNESS AND CATASTROPHIC FORGETTING — Sept 2020 - Nov 2020
Prof. Soheil Feizi, Dept. of Computer Science, University of Maryland

AWARDS/ACHIEVEMENTS

2024	Kulkarni Foundation Summer Research Fellowship (Declined)
2021 & 2023	CS Summer Research Fellowship, UMD
2020	Dean's Fellowship 2020, UMD
2019	Paper Selected for presentation at Samsung Best Paper Award (among top 8)
2019	Samsung Spot Award for Successful Implementation of Biopsy Needle Detection on Live Ultrasound Data
2019	Samsung Professional Level Software Certification
2013-14	Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship
2014	Among the top 0.3% of 1.3 million in the Joint Entrance Examination (JEE), India
2014	Letter of Appreciation from the HRD Minister for being in the top 1 percentile of CBSE (in English)

COURSEWORK

GRADUATE	• Advanced Techniques in Visual Learning and Recognition • Foundations of Deep Learning • AI and Existential Threats to Civilization • Advanced Numerical Optimization • Just Machine Learning • Computer Processing of Pictorial Information • Machine Learning Paradigms
UNDERGRADUATE	• Computer Vision using Machine Learning • Speech Processing • Artificial Intelligence • Parallel Algorithms • Computational Theory • Algorithmic Game Theory • Number Theory and Cryptography • Operating Systems • Software Engineering • Database Management System • Computer Networks • Probability Theory • Algorithms • Data Structures • Discrete Mathematics • Linear Algebra
ONLINE (Coursera)	1. Fundamentals of Digital Image and Video Processing by Aggelos K. Katsaggelos 2. Deep Learning Specialization by Andrew Ng

SKILLS

PROGRAMMING LANGUAGES	Python • C • C++
TOOLS, FRAMEWORKS, AND LIBRARIES	Pytorch • Keras • TensorFlow • ITK • Matlab*

** Elementary Proficiency*

[1] University of Maryland, College Park, [2] Samsung R&D Institute Bangalore, [3] Indian Institute of Technology Guwahati, [4] Indian Statistical Institute Kolkata
[5] Prime Videos & Amazon Studios, [6] Adobe